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Residual Denoising Diffusion Model for Seismic Fault Recognition Under Data Imbalance

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Summary

Accurate seismic fault identification is crucial for interpreting complex geological structures and efficiently exploring hydrocarbon resources. However, severe data imbalance in seismic datasets, especially in weak-signal regions and complex geological environments, significantly limits traditional fault recognition methods. To address this issue, this study introduces an innovative Residual Denoising Diffusion UNet (RDD-UNet), which integrates a lightweight 3D residual UNet architecture with a diffusion probabilistic model, reformulating fault detection as a conditional generative task to enhance accuracy and generalization under imbalanced data conditions.

The advantage of RDD-UNet lies in its residual denoising diffusion mechanism, accurately capturing the sparse spatial distribution of faults while precisely preserving critical boundary details. The model incorporates depthwise separable convolutions and residual connections, significantly improving computational efficiency for processing large-scale 3D seismic datasets.

Experiments involved initial training on synthetic seismic datasets, followed by rigorous evaluations on real-world F3-3D and Kerry-3D seismic surveys. Comparative results demonstrate that RDD-UNet surpasses traditional UNet, ResUNet, and basic diffusion models in terms of accuracy, noise resilience, and generalization, particularly excelling in detecting small-scale faults within complex geological settings. This research pioneers the application of residual denoising diffusion models in seismic fault recognition, providing an efficient and robust solution that significantly advances automated seismic interpretation technologies.

Keywords: seismic fault recognition, data imbalance, residual denoising diffusion model, deep learning, diffusion probabilistic model, small-scale fault detection

Introduction

Accurate seismic fault detection plays a crucial role in petroleum exploration and production, serving as a fundamental task for interpreting complex subsurface geological structures, identifying hydrocarbon migration paths, and defining reservoir traps (Wu et al. 2019). Traditional seismic interpretation methods largely depend on manual expertise, which is subjective, inefficient, and expensive given the ever-growing volume and complexity of seismic data. Therefore, automated, efficient, and accurate seismic fault interpretation techniques have become a critical research focus in recent years (An et al. 2023; Yang et al. 2022).

Recently, deep learning techniques, particularly convolutional neural networks (CNNs) and their derivatives such as UNet and ResUNet, have demonstrated substantial promise in seismic fault detection tasks (Ronneberger et al. 2015; Liao et al. 2024; Li et al. 2020; Yang et al. 2020). However, real-world seismic datasets commonly present several key challenges: (1) severe data imbalance, as fault pixels usually represent a very small portion of the entire seismic section, causing traditional models to be biased towards background classes and neglect weak fault signals (Monteiro et al. 2022; Monteiro et al. 2023); (2) difficulty in capturing weak-signal faults, where faint signals or thin faults are frequently obscured by noise, limiting the stability and sensitivity of conventional segmentation models (Dou et al. 2024; Zhu et al. 2021; Febriarta et al. 2024); (3) complexity from geological noise, such as salt domes and fault blocks, significantly reducing the model's generalization performance and increasing the difficulty of accurate detection.

To address these challenges, Diffusion Probabilistic Models (DDPMs) have recently exhibited outstanding capabilities in image generation tasks, especially in accurately depicting complex structures and fine details (Ho et al. 2020; Nichol and Dhariwal, 2021; Khader et al. 2023). Nevertheless, directly applying diffusion models to seismic data faces issues of blurriness and boundary detail loss (Jiang et al. 2023). Thus, this study introduces a novel Residual Denoising Diffusion UNet (RDD-UNet) model that synergizes the spatial feature extraction power of UNet with the generative capabilities of diffusion models. Additionally, the model innovatively incorporates a residual denoising mechanism (Liu et al. 2024; Jifara et al. 2019; He et al. 2016) and lightweight convolution (Wang et al. 2023; Yan et al. 2024) to significantly enhance its performance in detecting small-scale faults, weak signals, and boundary details, effectively improving accuracy and generalization under data-imbalanced conditions.

Methodology

The proposed Residual Denoising Diffusion UNet (RDD-UNet) model innovatively integrates diffusion probabilistic generative models with a residual UNet structure, significantly enhancing the accuracy and robustness of seismic fault recognition under imbalanced data conditions. The methodology comprises two main processes:

Forward Diffusion Process

The forward diffusion process progressively adds Gaussian noise to the original seismic fault label image x_0 , ultimately producing a purely noisy image x_T . This process is defined as a Markov chain at each step:

$$q(x_t | x_{t-1}) = N(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I) \quad (1)$$

The conditional probability transition $q(x_t | x_{t-1})$ from state x_{t-1} to x_t . By adjusting the noise factor β_t , the extent of noise introduced can be precisely controlled, effectively modeling the gradual corruption of data by noise.

Reverse Denoising Process

The reverse denoising process aims to reconstruct the original seismic fault label image from a noisy state. The RDD-UNet combines conditional diffusion modeling with a residual UNet structure to learn a parameterized conditional probability distribution:

$$P_{\theta}(x_{t-1}|x_t, c) = N(x_{t-1}; \mu_{\theta}(x_t, t, c), \sigma_t^2 I) \quad (2)$$

The conditional probability $p_{\theta}(x_{t-1}|x_t, c)$ predicted by the model to move from state x_{t-1} back to state x_t , where c represents seismic image conditioning information and θ are learnable parameters. Specifically, this conditional distribution is realized by predicting the noise term $\varepsilon_{\theta}(x_t, t, c)$:

$$\mu_{\theta}(x_t, t, c) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \varepsilon_{\theta}(x_t, t, c) \right) \quad (3)$$

The residual term $R(x_t)$, a novel residual structure designed in this study to enhance the recovery of boundary and detail information. The model employs a residual path $R(x_t)$ for further refinement of prediction results to improve reconstruction quality:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} (\varepsilon_{\theta}(x_t, t, c) + R(x_t)) \right) \quad (4)$$

The implementation of the noise predictor, where t is the diffusion timestep, and c is the conditional seismic input.

Training Objective

The core training objective is to minimize the discrepancy between the true noise and the predicted noise, represented as the loss function:

$$L(\theta) = E_{x_0, \epsilon} [\| \epsilon - \varepsilon_{\theta}(x_t, t, c) \|^2] \quad (5)$$

where E denotes the mathematical expectation and $\| \cdot \|^2$ indicates the L2 norm. Minimizing this loss allows the model to accurately capture fine seismic fault details and effectively handle data imbalance issues.

Through carefully designed forward diffusion and reverse denoising processes, the RDD-UNet model significantly improves the accuracy and robustness of seismic fault recognition, even under severely imbalanced seismic datasets.

Experimental Data

The datasets used in this study include one standard synthetic seismic fault dataset and three real-world seismic surveys. The synthetic dataset from [Wu et al. \(2019\)](#) contains diverse complex fault patterns suitable for initial model training and validation. Real-world datasets include the F3-3D survey from the North Sea (Netherlands) and Kerry-3D from offshore New Zealand, covering various geological features and noise conditions to validate model generalization.

For experiments, all data volumes were resized or padded to uniform dimensions of $128 \times 128 \times 128$ for efficient batch processing. Evaluation metrics include Accuracy, Intersection over Union (IoU), Precision, Recall, and F1-score. Comparisons were conducted against UNet, ResUNet, and traditional DDPM methods. During training, the AdamW optimizer was used with an initial learning rate of $1e-4$, a batch size of 4, and mixed-precision training implemented on a single 4090 GPU device.

Application and Results

Experimental evaluations demonstrate that the RDD-UNet achieves an average Precision of 0.9383 on synthetic dataset, significantly outperforming UNet (0.8642), ResUNet (0.9169), and standard denoising diffusion models (0.9214). RDD-UNet achieves an IoU of 0.7287 and an F1 Score of 0.8419, compared to DD-UNet's IoU of 0.6957 and F1 Score of 0.8193. These improvements highlight the model's ability to effectively capture detailed fault features while managing imbalanced fault distributions. Moreover, RDD-UNet shows stronger robustness in detecting small-scale faults, weak signals, and under complex geological conditions. Specifically, the missed detection rate for thin-layer and weak-signal faults in Kerry-3D dataset was reduced by approximately 13%. Regarding inference speed, RDD-UNet demonstrates about a 16% improvement over ResUNet with fewer parameters, confirming the effectiveness of its lightweight architecture.

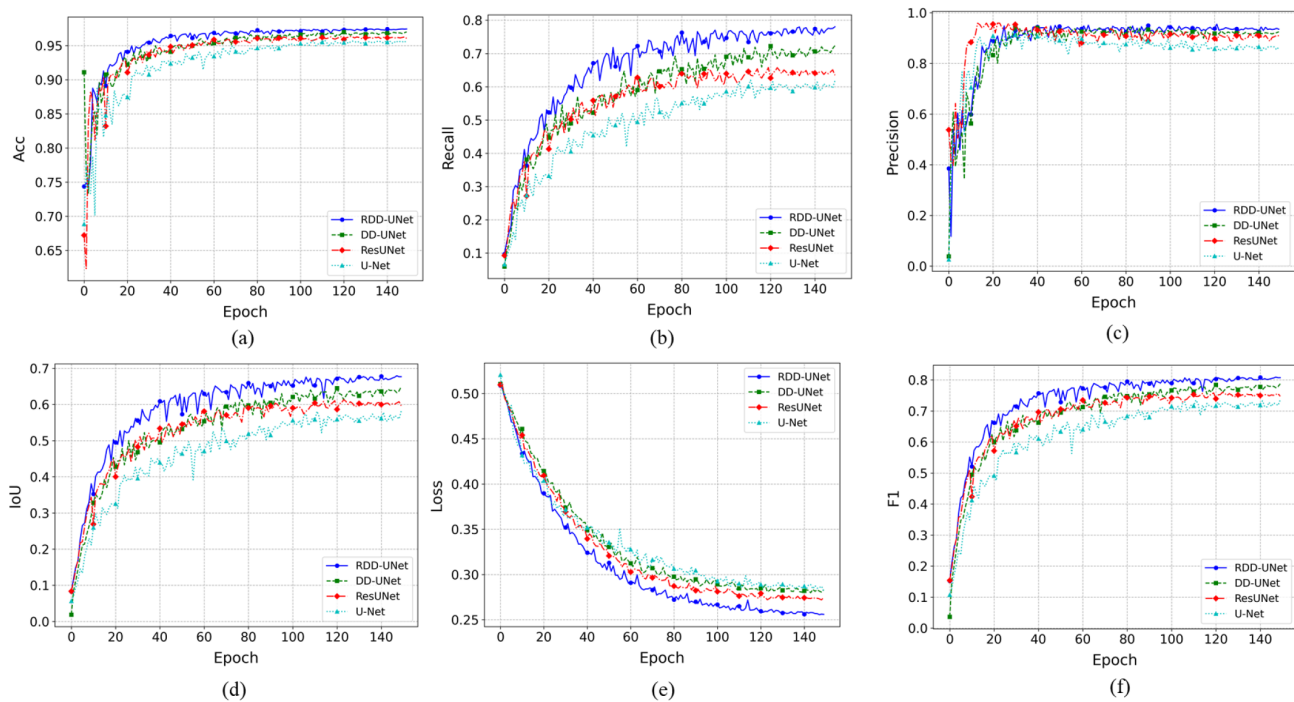


Figure 1—Evaluation curves. (a) Acc; (b) Recall; (c) Precision; (d) IoU; (e) Loss; (f) F1 score. Each curve corresponds to a different segmentation model, U-Net: cyan line, ResUNet: red line, DD-UNet: green line, RDD-UNet: blue line.

In the numerical experiments, we compared the performance of U-Net, ResUNet, DD-UNet, and the proposed RDD-UNet model on real-world seismic datasets for fault interpretation.

Fig. 2 shows the fault detection results of each model on the F3-3D field dataset using a subset volume of size $512 \times 384 \times 128$. From the visualization, it is evident that the fault probability maps predicted by RDD-UNet exhibit significantly better coherence and boundary sharpness. The fault structures are more continuous and well-defined, and the influence of background noise is notably reduced. In contrast, U-Net and ResUNet produce fault edges that are blurry and lack clear transitions, while DD-UNet struggles to preserve structural integrity, often resulting in discontinuous faults and pseudo-fault artifacts.

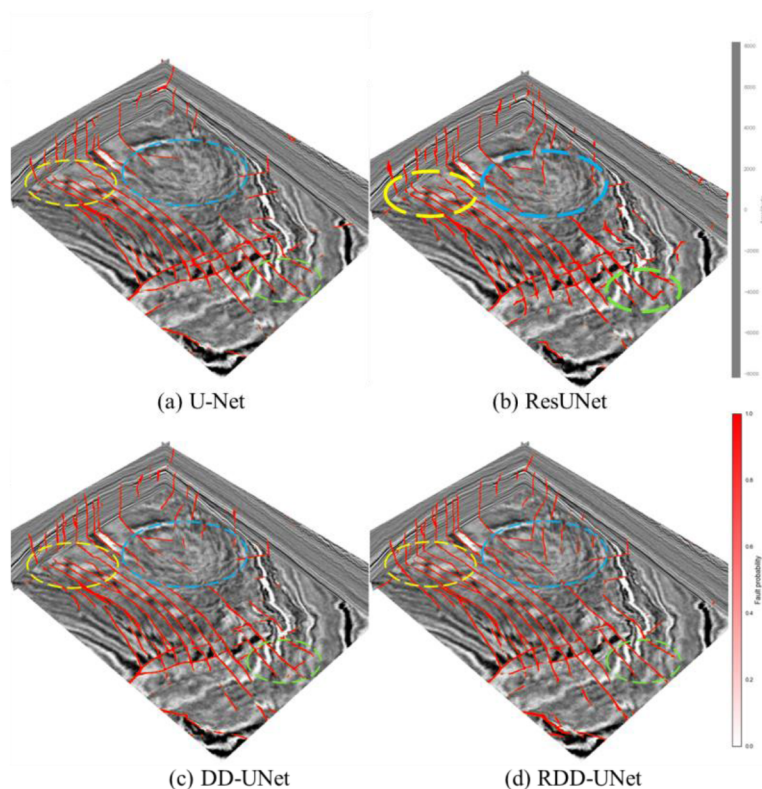


Figure 2—Fault identification results on the F3-3D field data. (a) U-Net; (b) ResUNet; (c) DD-UNet; (d) RDD-UNet. The red lines represent the predicted fault structures, with the blue rectangle highlighting regions of densely overlapping faults, while the yellow rectangle emphasizes areas of complex fault intersections. The results demonstrate that RDD-UNet exhibits significant advantages in fault continuity and structural integrity. The gray color bar at the top right represents the amplitude range of the seismic data, while the red color bar at the bottom right indicates the probability of fault occurrence, ranging from 0 (no fault) to 1 (high fault), providing an intuitive display of the fault prediction intensity and distribution.

Fig. 3 further illustrates the results on the more geologically complex Kerry-3D field dataset (subset size: $156 \times 432 \times 160$), posing greater challenges to all models. Even in this challenging setting, RDD-UNet demonstrates superior capability in preserving structural consistency, maintaining coherent fault orientations, and capturing fine-scale fault details. Additionally, it maintains strong boundary localization and robustness against geological noise. Other models, however, exhibit more pronounced discontinuities, loss of detail, and increased interference, leading to suboptimal interpretation performance.

These results comprehensively validate the robustness and accuracy of RDD-UNet across seismic datasets with varying levels of geological complexity.

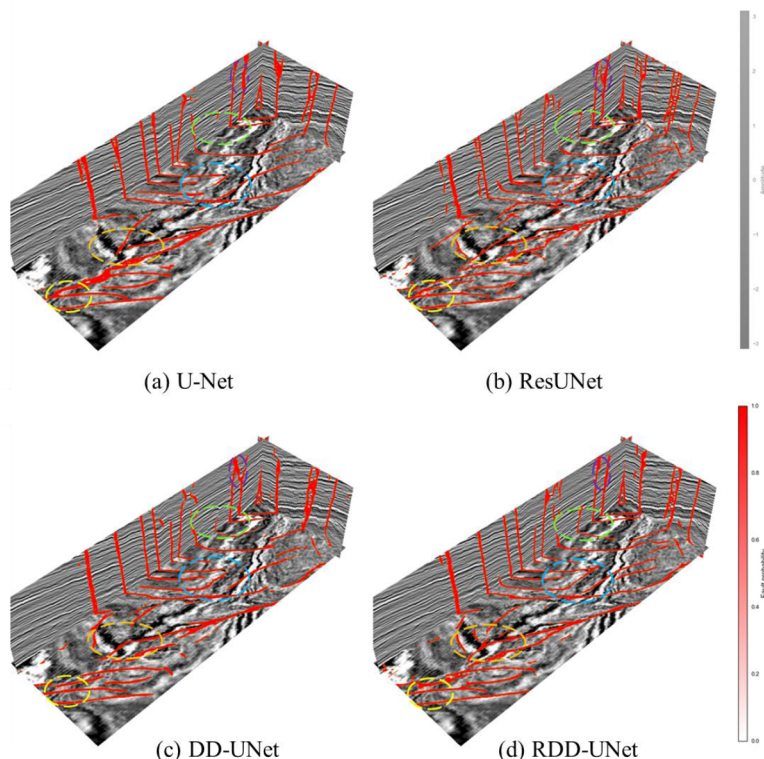


Figure 3—Fault identification results on the Kerry-3D field data. (a) U-Net; (b) ResUNet; (c) DD-UNet; (d) RDD-UNet. The red lines indicate the predicted fault structures. The blue rectangle highlights densely overlapping fault regions, the yellow rectangle emphasizes complex fault intersection areas, and the green rectangle focuses on regions with subtle fault features. The gray color bar on the right represents the amplitude range of the seismic data, while the red color bar represents fault probability, ranging from 0 (no fault) to 1 (high fault). The results demonstrate that RDD-UNet achieves superior fault continuity and structural integrity in complex fault scenarios.

Conclusions

In this study, we propose a novel Residual Denoising Diffusion Network Model (RDD-UNet) to tackle the critical challenges of data imbalance and weak-signal fault detection in seismic interpretation. By seamlessly integrating a lightweight residual UNet with a diffusion probabilistic framework, the model effectively enhances both the generalization ability and robustness to noise in complex geological settings. Experimental results demonstrate that RDD-UNet consistently outperforms conventional U-Net, ResUNet, and DD-UNet, especially in detecting small-scale faults and subtle features within noisy seismic volumes.

The model's strength lies in its residual denoising mechanism and the use of depthwise separable convolutions, which enable it to capture sparse fault geometries while preserving fine boundary details with high computational efficiency. This makes the model not only accurate but also highly deployable for large-scale, real-time 3D seismic interpretation tasks.

Several directions for future research emerge from this work: (1) Uncertainty quantification could be incorporated to assist geophysicists in assessing the reliability of fault predictions; (2) Cross-domain generalization and transfer learning approaches may enhance the adaptability of the model to diverse geological settings; (3) Integration with velocity model building workflows could enable a more holistic automated interpretation pipeline; (4) Performance assessment under varying signal-to-noise ratios may further validate the model's robustness in challenging environments.

From the conducted research, we summarize the following key conclusions:

- RDD-UNet combines the strengths of denoising diffusion processes and deep segmentation architectures, allowing accurate detection of complex and small-scale faults even under extreme data imbalance conditions, where weak signals are often buried in noise.
- Compared to U-Net, ResUNet, and DD-UNet, RDD-UNet achieves superior accuracy, spatial continuity, and boundary delineation across various real-world seismic datasets—particularly in geologically complex areas such as intersecting faults and low-SNR regions.
- The use of lightweight convolutional modules and residual denoising blocks ensures high recognition precision while maintaining a low parameter count and computational burden, enhancing the model's practical deployability in industrial applications.
- By incorporating prior knowledge of fault geometries into the spatial generation path, RDD-UNet improves prediction stability in uncertain regions, laying a foundation for more robust 3D automated seismic interpretation systems.
- The model's design is extensible, offering potential applications in fault attribute prediction, multi-scale strike-slip detection, and integration into end-to-end inversion-interpretation workflows.

In conclusion, RDD-UNet not only advances the theoretical application of diffusion models in geophysical imaging but also delivers a high-performance and scalable solution for seismic fault interpretation in real-world scenarios. Future research may further explore its synergistic use with velocity modeling and stratigraphic property analysis to enable fully intelligent and automated seismic exploration.

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